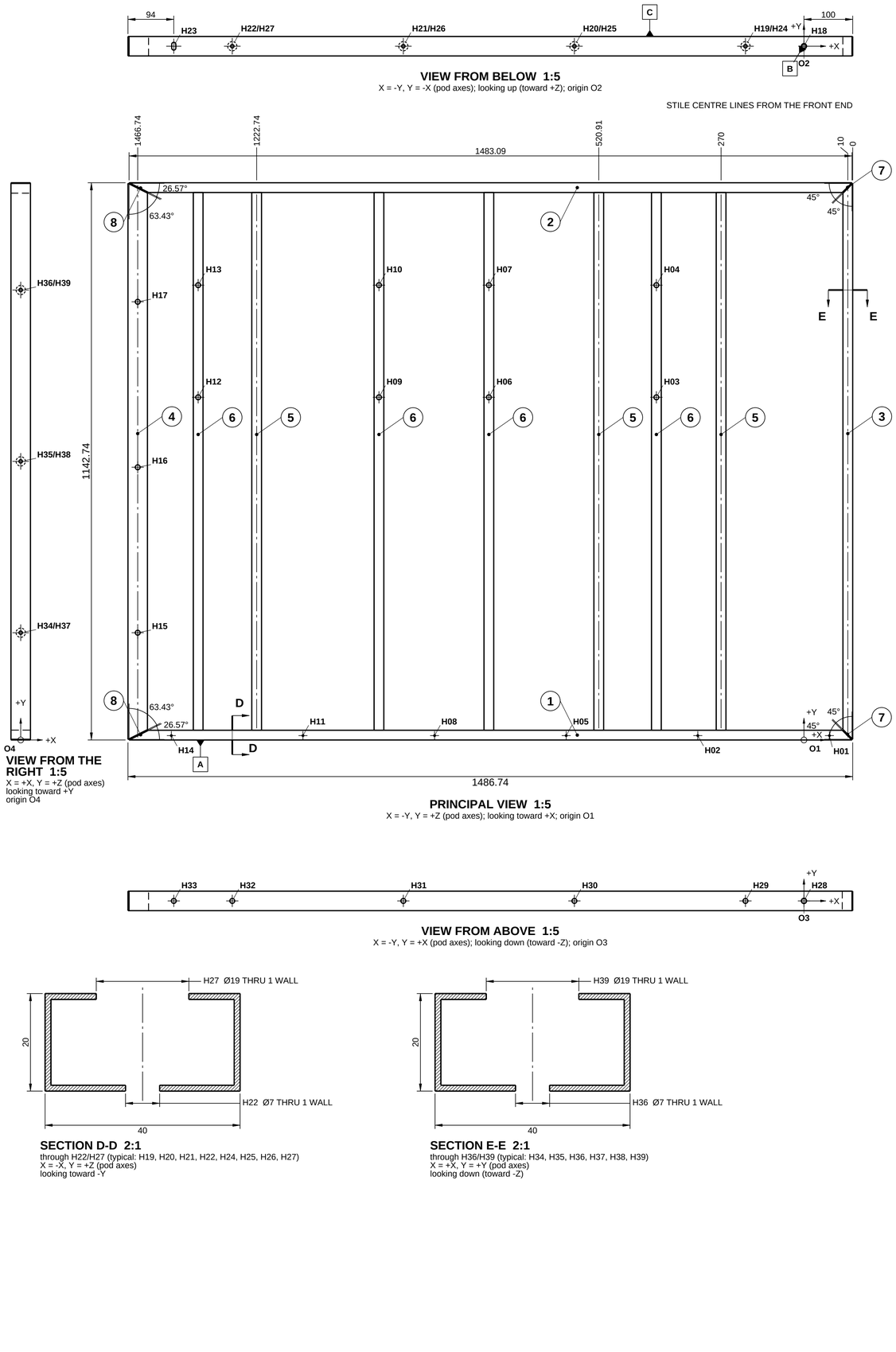


VIEW FROM THE RIGHT 1:5
X = +X, Y = +Z (pod axes)
looking toward +Y
origin O4

SECTION D-D 2:1
through H22/H27 (typical: H19, H20, H21, H22, H24, H25, H26, H27)
X = -X, Y = +Z (pod axes)
looking toward -Y

SECTION E-E 2:1
through H36/H39 (typical: H34, H35, H36, H37, H38, H39)
X = +X, Y = +Y (pod axes)
looking down (toward -Z)

HOLES H01-H17: 212201 WALL - FRAME RAIL BOTTOM - SIDE LOWER - RH + 212204 WALL - FRAME STILE BACK 40 X 40 - SIDE LOWER - RH + 212206 WALL - FRAME STUD WITH LEDGE BRACKET RIVNUTS - SIDE LOWER - RH, left face, room side
seen in the principal view; origin O1; +X right, +Y up; drilled from this face

TAG	X (mm)	Y (mm)	SIZE	DEPTH	FEATURE (register) AND FITS
H01	52.26	9.00	Ø3.3	THRU 1 WALL (1.2)	F50 Cassette ledge rivet holes (838012); rivet hole: blind rivet 838012, closed end, countersunk, Ø3.2 (ISO 15974)
H02	-217.74	9.00	Ø3.3	THRU 1 WALL (1.2)	F50 Cassette ledge rivet holes (838012); rivet hole: blind rivet 838012, closed end, countersunk, Ø3.2 (ISO 15974)
H03	-302.74	702.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H04	-302.74	932.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H05	-487.74	9.00	Ø3.3	THRU 1 WALL (1.2)	F50 Cassette ledge rivet holes (838012); rivet hole: blind rivet 838012, closed end, countersunk, Ø3.2 (ISO 15974)
H06	-646.37	702.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H07	-646.37	932.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H08	-757.74	9.00	Ø3.3	THRU 1 WALL (1.2)	F50 Cassette ledge rivet holes (838012); rivet hole: blind rivet 838012, closed end, countersunk, Ø3.2 (ISO 15974)
H09	-871.83	702.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H10	-871.83	932.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H11	-1027.74	9.00	Ø3.3	THRU 1 WALL (1.2)	F50 Cassette ledge rivet holes (838012); rivet hole: blind rivet 838012, closed end, countersunk, Ø3.2 (ISO 15974)
H12	-1242.74	702.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H13	-1242.74	932.74	Ø10	THRU 1 WALL (0.8)	F54 INTERIOR's fixing rivnuts (838003)
H14	-1297.74	9.00	Ø3.3	THRU 1 WALL (1.2)	F50 Cassette ledge rivet holes (838012); rivet hole: blind rivet 838012, closed end, countersunk, Ø3.2 (ISO 15974)
H15	-1366.74	220.00	Ø10	THRU 1 WALL (1)	F09 J-WALL-02 corner rivnuts, nut side
H16	-1366.74	559.30	Ø10	THRU 1 WALL (1)	F09 J-WALL-02 corner rivnuts, nut side
H17	-1366.74	898.61	Ø10	THRU 1 WALL (1)	F09 J-WALL-02 corner rivnuts, nut side

HOLES H18-H23: 212201 WALL - FRAME RAIL BOTTOM - SIDE LOWER - RH, bottom face
seen in the view from below; origin O2; +X right, +Y up; drilled from this face

TAG	X (mm)	Y (mm)	SIZE	DEPTH	FEATURE (register) AND FITS
H18	0.00	0.00	Ø8.5	THRU 1 WALL (1.2)	F03 Locating
H19	-120.00	0.00	Ø7	THRU 1 WALL (1.2)	F01 J-BASE-01 bolt clearance, bolt side (838001, 838004, 838003)
H20	-470.91	0.00	Ø7	THRU 1 WALL (1.2)	F01 J-BASE-01 bolt clearance, bolt side (838001, 838004, 838003)
H21	-821.83	0.00	Ø7	THRU 1 WALL (1.2)	F01 J-BASE-01 bolt clearance, bolt side (838001, 838004, 838003)
H22	-1172.74	0.00	Ø7	THRU 1 WALL (1.2)	F01 J-BASE-01 bolt clearance, bolt side (838001, 838004, 838003)
H23	-1292.74	0.00	SLOT 8.5 x 16 OVERALL, R4.25 ENDS along Y	THRU 1 WALL (1.2)	F03 Locating

HOLES H24-H27: 212201 WALL - FRAME RAIL BOTTOM - SIDE LOWER - RH, top face (inside the member's bay)
far wall, coaxial with H19-H22: hidden in the view from below, shown in section D-D; origin O2; +X right, +Y up; drilled from this face

TAG	X (mm)	Y (mm)	SIZE	DEPTH	FEATURE (register) AND FITS
H24	-120.00	0.00	Ø19	THRU 1 WALL (1.2)	F02 Key access for F01 (108001)
H25	-470.91	0.00	Ø19	THRU 1 WALL (1.2)	F02 Key access for F01 (108001)
H26	-821.83	0.00	Ø19	THRU 1 WALL (1.2)	F02 Key access for F01 (108001)
H27	-1172.74	0.00	Ø19	THRU 1 WALL (1.2)	F02 Key access for F01 (108001)

HOLES H28-H33: 212202 WALL - FRAME RAIL TOP - SIDE LOWER - RH, top face
seen in the view from above; origin O3; +X right, +Y up; drilled from this face

TAG	X (mm)	Y (mm)	SIZE	DEPTH	FEATURE (register) AND FITS
H28	0.00	0.00	Ø10	THRU 1 WALL (1.2)	F05 J-WALL-01 rivnuts, nut side (4, one per bay) and 2 spigot rivnuts
H29	-120.00	0.00	Ø10	THRU 1 WALL (1.2)	F05 J-WALL-01 rivnuts, nut side (4, one per bay) and 2 spigot rivnuts
H30	-470.91	0.00	Ø10	THRU 1 WALL (1.2)	F05 J-WALL-01 rivnuts, nut side (4, one per bay) and 2 spigot rivnuts
H31	-821.83	0.00	Ø10	THRU 1 WALL (1.2)	F05 J-WALL-01 rivnuts, nut side (4, one per bay) and 2 spigot rivnuts
H32	-1172.74	0.00	Ø10	THRU 1 WALL (1.2)	F05 J-WALL-01 rivnuts, nut side (4, one per bay) and 2 spigot rivnuts
H33	-1292.74	0.00	Ø10	THRU 1 WALL (1.2)	F05 J-WALL-01 rivnuts, nut side (4, one per bay) and 2 spigot rivnuts

HOLES H34-H36: 212203 WALL - FRAME STILE FRONT - SIDE LOWER - RH, front face
seen in the view from the right; origin O4; +X right, +Y up; drilled from this face

TAG	X (mm)	Y (mm)	SIZE	DEPTH	FEATURE (register) AND FITS
H34	0.00	220.00	Ø7	THRU 1 WALL (1.2)	F07 J-FRONT-01 bolt clearance, bolt side
H35	0.00	571.37	Ø7	THRU 1 WALL (1.2)	F07 J-FRONT-01 bolt clearance, bolt side
H36	0.00	922.74	Ø7	THRU 1 WALL (1.2)	F07 J-FRONT-01 bolt clearance, bolt side

HOLES H37-H39: 212203 WALL - FRAME STILE FRONT - SIDE LOWER - RH, back face (inside the member's bay)
far wall, coaxial with H34-H36: hidden in the view from the right, shown in section E-E; origin O4; +X right, +Y up; drilled from this face

TAG	X (mm)	Y (mm)	SIZE	DEPTH	FEATURE (register) AND FITS
H37	0.00	220.00	Ø19	THRU 1 WALL (1.2)	F08 Key access for F07, in the stile's back wall facing the first bay
H38	0.00	571.37	Ø19	THRU 1 WALL (1.2)	F08 Key access for F07, in the stile's back wall facing the first bay
H39	0.00	922.74	Ø19	THRU 1 WALL (1.2)	F08 Key access for F07, in the stile's back wall facing the first bay

PROJECT: **Spaceship (FocusPod v2)**	STATUS: **DRAFT**	DATE: 2026-11-02	SHEET: **1/2**

TITLE
WALL - PANEL FRAME - SIDE LOWER - RH
Vách phải - khung dưới

PART NUMBER	REV	DRAWING NUMBER (FILE)
211102	**P01**	**211102_P01_WALL-PANEL-FRAME-SIDE-LOWER-RH_DWG**

MATERIAL
MATERIAL: the steel box tube members below and the 4 mitre plates, welded (D-131)

FINISH
FINISH: pre-galvanized tube, zinc Z100; every weld and burnt zone wire-brushed and coated with a 2-component zinc-rich epoxy; no hot-dip (J-FRAME-01); the skin's bond prep per the bond spec

SCALE	SIZE	UNITS	MASS
1:5 (sections D-D E-E 2:1)	A1	**UNITS: mm**	11.455 kg (calculation)

CREATED BY	APPROVED	TOLERANCES
Autonomous	not released	TOLERANCE: see the notes; not ISO 2768

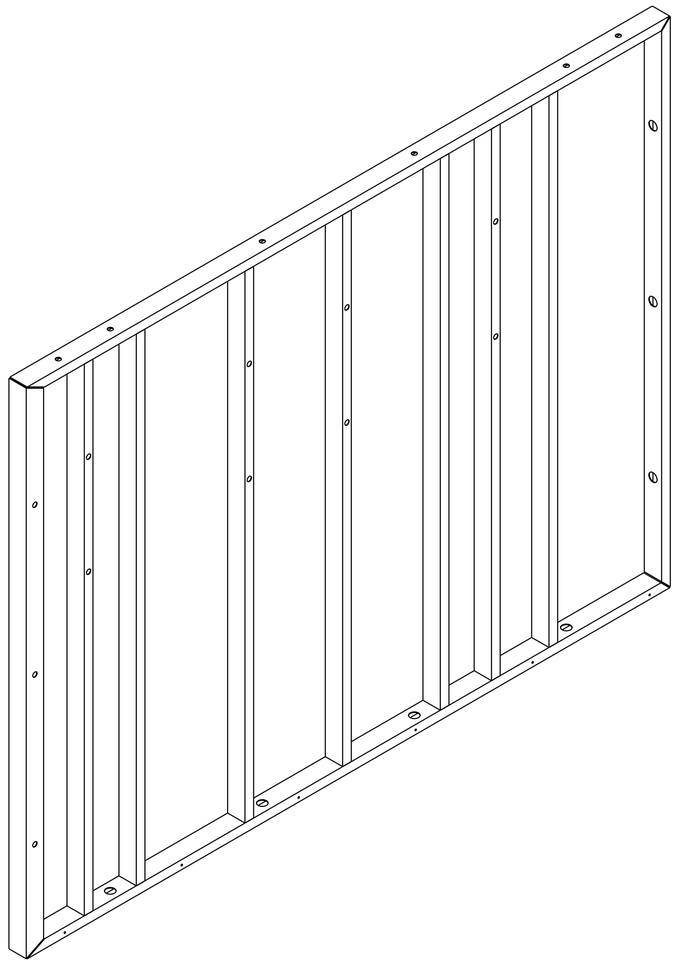
PROJECTION: ISO first angle (ISO 5456-2)

GENERAL NOTES

1. UNITS: mm. Do not scale the drawing. ISO first-angle projection (ISO 5456-2), symbol in the title block. Dimensions to ISO 129-1. Sheet 1: views, sections and hole tables; sheet 2: these general notes, the cut list.
2. Views: the principal view looks toward +X (X = -Y, Y = +Z (pod axes); looking toward +X); every view's title gives its own X and Y and how they map to the pod axes (conventions section 4). On every view X runs right and Y up.
3. Datums (ISO 5459): A = bottom face of 212201 WALL - FRAME RAIL BOTTOM - SIDE LOWER - RH (the seat on the band); B = axis of the spigot hole H18 (joints register R-J6, R-J13); C = the room-side face (X 1204), which stops the turn about B (the spigot slot H23 runs across the rail, R-J17, so it locates along the rail only). Origin O = datum B's axis on datum A; O1, O2... are O seen in each view. Hole positions in the tables are theoretically exact from O.
4. FRAME: welded, galvanized steel box tube (D-131, J-FRAME-01). Corners mitred, with a 2.0 mm mitre plate welded into each mitre (831007 to 831009, placed as the joints CAD README gives). Every corner and T joint welded all round, continuously, so each tube stays sealed. MIG brazing with CuSi3 wire (ISO 24373 S Cu 6560) under argon preferred; or short-arc MAG with G3Si1 wire (ISO 14341-A) if the factory's welding procedure passes J-T7's coupons (HOLD-22).
5. Weld in a jig that holds every member flat and square on its skin side: diagonally opposite corners in turn, then the T joints from the middle out, alternating sides; the frame cools in the jig. Beads on the skin-side faces and the seam faces ground flush (0 to 0.2 proud, never more than 0.3 below the face).
6. Pre-galvanized tube, zinc Z100: no hot-dip galvanizing after welding, so no vent or drain holes. Every burnt zone wire-brushed and coated with a 2-component zinc-rich epoxy (about 40 µm dry; assumption).
7. Every joint hole is made after welding and the touch-up, by CNC or a drill jig that locates on the frame's joint faces and spigot hole; rivnuts 838003 (zinc-plated steel) are set after the touch-up.
8. Holes go THRU 1 WALL of the member unless the table says otherwise; deburr both sides. Coaxial holes in two walls of one member are shown in sections D-D and E-E.
9. TOLERANCES (PROPOSED until the shop confirms; not ISO 2768): hole and slot positions Ø0.4 to A|B|C (R-J6); spigot hole and slot ±0.1 from the frame ends (joints register section 4); rivnut holes Ø10.00 +0.10/0 (D-163, Sherex CAL2-610-4.2); other hole sizes +0.2/0 (assumption); cut lengths ±0.5 (parts list); welded frame: seam faces flat within 0.5, skin-side faces within ±0.5 of one plane lying free, diagonals equal within 1.5, measured on a flat table after welding, before the holes; straighten cold or reject, never clamp flat (J-FRAME-01).
10. Source: v2/03-Manufacturing/wall/211102_P01_WALL-PANEL-FRAME-SIDE-LOWER-RH_STEP.step (sha256 1b80226793c2; declared unit cm, read in mm and checked against tools/checks/stepfile.py, worst box difference 0 mm); parts list v2/03-Manufacturing/wall/210102_P01_WALL-PANEL-ASSY-SIDE-LOWER-RH_BOM.csv; feature rationale v2/02-Drawings/wall/FEATURE-RATIONALE-210102.md. From that STEP: regenerate it, never edit the PDF.

CUT LIST (cut list 210000_P01_WALL-TUBE-CUT-LIST_BOM.csv)
the steel box tube members below and the 4 mitre plates, welded (D-131)

ITEM	MEMBER	PROFILE (mm)	CUT LENGTH (mm)	QTY	ENDS, TOLERANCE
1	212201 WALL - FRAME RAIL BOTTOM - SIDE LOWER - RH	steel box tube 20 x 40 x 1.2, S280GD+Z (EN 10346) or SGC400 (JIS G3302): certified yield 280 MPa or more, zinc Z100 (J-FRAME-01, D-131)	1483.09	1	front end mitred 45.00° and back end 63.43° off square, in the frame's plane, long points on its bottom face (the frame's outer edge); cut back for the 2 mm mitre plate (D-117, D-131); length ±0.5 (parts list)
2	212202 WALL - FRAME RAIL TOP - SIDE LOWER - RH	steel box tube 20 x 40 x 1.2, S280GD+Z (EN 10346) or SGC400 (JIS G3302): certified yield 280 MPa or more, zinc Z100 (J-FRAME-01, D-131)	1483.09	1	front end mitred 45.00° and back end 63.43° off square, in the frame's plane, long points on its top face (the frame's outer edge); cut back for the 2 mm mitre plate (D-117, D-131); length ±0.5 (parts list)
3	212203 WALL - FRAME STILE FRONT - SIDE LOWER - RH	steel box tube 20 x 40 x 1.2, S280GD+Z (EN 10346) or SGC400 (JIS G3302): certified yield 280 MPa or more, zinc Z100 (J-FRAME-01, D-131)	1139.92	1	bottom and top ends both mitred 45.00° off square, in the frame's plane, long points on its front face (the frame's outer edge); cut back for the 2 mm mitre plate (D-117, D-131); length ±0.5 (parts list)
4	212204 WALL - FRAME STILE BACK 40 X 40 - SIDE LOWER - RH	steel box tube 40 x 40 x 1, S280GD+Z (EN 10346) or SGC400 (JIS G3302): certified yield 280 MPa or more, zinc Z100 (J-FRAME-01, D-131)	1140.5	1	bottom and top ends both mitred 26.57° off square, in the frame's plane, long points on its back face (the frame's outer edge); cut back for the 2 mm mitre plate (D-117, D-131); length ±0.5 (parts list)
5	212205 WALL - FRAME STILE MID - SIDE LOWER - RH	steel box tube 20 x 40 x 0.8, S280GD+Z (EN 10346) or SGC400 (JIS G3302): certified yield 280 MPa or more, zinc Z100 (J-FRAME-01, D-131)	1102.74	3	square-cut ends; length ±0.5 (parts list)
6	212206 WALL - FRAME STUD WITH LEDGE BRACKET RIVNUTS - SIDE LOWER - RH	steel box tube 20 x 40 x 1.2, S280GD+Z (EN 10346) or SGC400 (JIS G3302): certified yield 280 MPa or more, zinc Z100 (J-FRAME-01, D-131)	1102.74	4	square-cut ends; length ±0.5 (parts list)
7	831007 HARDWARE - MITRE PLATE - 20 X 40 - 45 DEG - 2.0 STEEL	steel plate 2.0, galvanized Z100, yield 280 MPa or more (as the frame tube, J-FRAME-01)	-	2	welded in; no drawing of its own, its parts-list row defines it (D-125)
8	831008 HARDWARE - MITRE PLATE - 20 X 40 TO 40 X 40 - 2.0 STEEL	steel plate 2.0, galvanized Z100, yield 280 MPa or more (as the frame tube, J-FRAME-01)	-	2	welded in; no drawing of its own, its parts-list row defines it (D-125)



ISOMETRIC VIEW 1:5
for identification only

PROJECT: **Spaceship (FocusPod v2)**	STATUS: **DRAFT**	DATE: 2026-11-02	SHEET: **2/2**

TITLE
WALL - PANEL FRAME - SIDE LOWER - RH
Vách phải - khung dưới

PART NUMBER	REV	DRAWING NUMBER (FILE)
211102	**P01**	**211102_P01_WALL-PANEL-FRAME-SIDE-LOWER-RH_DWG**

MATERIAL
MATERIAL: the steel box tube members below and the 4 mitre plates, welded (D-131)

FINISH
FINISH: pre-galvanized tube, zinc Z100; every weld and burnt zone wire-brushed and coated with a 2-component zinc-rich epoxy; no hot-dip (J-FRAME-01); the skin's bond prep per the bond spec

SCALE	SIZE	UNITS	MASS
1:5 (sections D-D E-E 2:1)	A1	**UNITS: mm**	11.455 kg (calculation)

CREATED BY	APPROVED	TOLERANCES
Autonomous	not released	TOLERANCE: see the notes; not ISO 2768

PROJECTION: ISO first angle (ISO 5456-2)